



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH Dual Degree 5Y	IIT Kharagpur	6.71 / 10
2021	AISSCE	Jawahar Navodaya Vidyalaya,Kottayam,Kerala	91.6%
2019	AISSE	Jawahar Navodaya Vidyalaya,Purulia,West Bengal	95%

INTERNSHIPS

PRISM (OYO - USA/UK) Data Science Intern (Worked on various projects, including real-time pricing dashboards , review analysis , call centre analytics , pricing debuggers , error reporting and chatbots)	[May 2026 - October 2026]
- Real-time pricing app (FastAPI, Docker, Panel) across 250+ UK hotels with live A/B testing, occupancy and competitor analytics for RMs LangChain/LangGraph review dashboard for 2,000+ US/UK properties: sentiment analysis, label tagging and summarization on Kubernetes - Engineered LangGraph and OpenAI multi-agent debuggers to solve pricing escalations, achieving 80% accuracy in automated resolutions. - Built Linux cron-triggered logs and instant error alerts via SMTP and data visualizations, ensuring 99.9% continuous system uptime.	
Brix AI Technology LLP Gen AI and ML Development Intern (Engineered scalable NLP pipelines, deployed generative RAG chatbots, and automated US-RCM analytics and fine-tuned speech classification models)	[May 2025 - July 2026]
- Developed &integrated a customer support chatbot leveraging Llama, RAG, Fastapi and sentence-transformers for embedding extraction - Engineered NLP data pipelines for real-time US RCM call analysis, structuring scalable MongoDB data models; deployed predictive analytics and automated reporting algorithms, extracting key actionable business insights and accelerating data processing throughput by 40% - Fine-tuned Wav2Vec2 using Transformer model for US vs Indian accent classification, improving speech model accuracy to 94.2% overall.	
Proficient Vision Solutions Pvt. Ltd. Computer Vision Intern (Deployed AWS EC2 video pipelines using PyTorch, YOLOv8, and PaddleOCR for automated traffic analytics, achieving 95% spatial filtering)	[November 2025 - January 2026]
- Engineered an AWS EC2 video processing pipeline using PyTorch and OpenCV, implementing spatial filtering for 95% fog/rain removal. - Deployed AWS EC2 video pipelines with custom NGINX RTMP, achieving a 40% PSNR increase in image clarity at 30+ FPS. Integrated YOLOv8 (92% mAP) and PaddleOCR (98% accuracy) to log 10K+ license plates into automated CSVs for real-time traffic analytics systems - Built custom NGINX RTMP servers replacing AWS MediaLive and MediaPackage, streaming video data to achieve 8-second latency.	
Centre for Robotics (IIT Kharagpur) Research Intern (Engineered optimized deep learning and computer vision models to power autonomous tomato harvesting for agricultural robotics applications)	[May 2024 - July 2024]
- Benchmarked Ultralytics YOLO against DETR on COCO, recording 53.9% versus 42.0% mAP, and trained both models on Roboflow's tomato dataset for real-time object detection, segmentation, and classification, achieving 91.5% validation mAP at 45 FPS using PyTorch framework. - Accelerated object detection using TensorRT and CUDA, achieving 1.68-2.33x (Python) and 3.24-3.89x (C++) faster inference than PyTorch. - Worked with PyRealSense depth cameras, inverse kinematics, and OpenCV to map 3D spatial coordinates for robotic path manipulation.	

PROJECTS

Autonomous Multi-Agent Research & Reasoning System B.Tech Project-I	[July 2025 - November 2025]
- Architected LangGraph multi-agent pipeline (planner, retriever, critic, synthesizer) with ReAct + Reflexion loops achieving 87% accuracy. - Engineered hybrid RAG with FAISS dense retrieval, BM25 sparse search, and HyDE reranking, reducing hallucinations by 41% - Deployed async FastAPI with LangChain tool-calling, Redis session memory, and LangSmith tracing achieving <2.1s median latency - Implemented RAGAS evaluation (context recall 0.91, answer relevancy 0.88) with Grafana enabling feedback-driven prompt refinement.	
Attention Based Image Captioning using CNN-RNN Pipeline B.Tech Project-II	[January 2025 - April 2026]
EfficientNetV2 binary classifier (12K dataset, 98.7% accuracy) gates images to two Transformer captioning branches at 0.06s inference. - NITC: EffNetV2 + Transformer decoder (8-head, d=512, AdamW); 65.8 BLEU-1 / 48.3 METEOR on Flickr8k, +8.2 over CNN-LSTM baseline - GIBC: EasyOCR (98.2% accuracy) extracts chart text; VGG16 + OCR embeddings fused in BiTransformer decoder for grounded captions. - Label-smoothed CE (=0.1) + gradient clipping; 34% fewer hallucinations; FastAPI/TorchScript at 0.31s/image with Grad-CAM attention.	

SKILLS AND EXPERTISE

Languages: Python | C++ | C | HTML | SQL **Tools:** Google Colab | Visual Studios Code | Conda | Chroma DB | Jupyter Notebooks | GitHub | MS Office | Git | Overleaf (LaTeX) | Parquet (pyarrow) | CSV/JSON | Docker | Linux | Nginex | LiteLLM (* - elementary proficiency)
Libraries: Numpy | Pandas | Tensorflow / Keras | PyTorch | Seaborn | Matplotlib | Scikit-learn | OpenCV | SciPy | Transformers | Langchain | langgraph | NLTK | SpykeTorch | SciBERT | tokenizers | sentence-transformers | imbalanced-learn | Fastapi | Panel | Redish | Geopy

COURSEWORK INFORMATION

CS/AI: Programming and Data Structures | Algorithms | Computer Architecture | Pattern Recognition and Machine Intelligence | Deep Learning | Advanced Operating Systems Design | Probability and Statistics | Linear Algebra and Optimization Models | Advanced Calculus,

EXTRA CURRICULAR ACTIVITIES

- Represented the Dakshana India Educational Trust Fund at JNV Bankura, successfully conducting the NDST (2026) exam for 113 students.
- Served as NSS Group Leader (1st+2nd yr), executing cleaning, teaching, awareness drives, and health camps in our assigned nearby village.